Deep Learning-Aided Phase-Aware Channel Estimation for Error-Floor Suppression over Phase-Drifting Backscatter Links

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***Abstract*—This paper proposes a deep learning-aided phase-aware channel estimation (CE) method for monostatic backscatter links that suffer from a slowly drifting carrier phase across the pilot block caused by residual carrier-frequency offset (CFO) and reader-side synthesizer mismatch after coarse synchronization. Closed-form model-based estimators address this drift by linearizing the phase-slope-induced rotation across the pilot block, which is accurate only within a limited drift range; beyond it, the residual bias becomes independent of noise and the estimation error saturates at a deterministic floor. The closed-form estimate is retained as a model-based anchor, and a residual-gated convolutional neural network (RG-CNN) applies a bounded correction that stays inactive while the linearized model holds and acts only under large drift, preserving model-based accuracy in the small-drift regime. Numerical results show that the proposed estimator avoids the error-floor saturation of the model-based estimate and reduces both the CE error and the large-drift failure probability.**

***Index Terms*—Backscatter communications, deep learning, convolutional neural network, channel estimation.**

I. Introduction

Backscatter communication has attracted attention as an enabling technology for low-power Internet-of-Things (IoT) connectivity. By conveying data through reflection of an incident carrier instead of generating one, passive tags remove the active radio-frequency front end and operate at extremely low power, which suits massive, energy-constrained deployments [1]. A practical limitation of such links is reliable channel estimation (CE) under front-end impairments: in the monostatic configuration, direct carrier leakage introduces a baseband offset, while residual carrier-frequency offset (CFO) and frequency synthesizer mismatch leave a slowly drifting carrier phase across the pilot block after coarse synchronization. This residual phase drift becomes a dominant impairment for CE, as it acts multiplicatively on the observation and biases the estimate of the cascaded backscatter channel [2].

Deep learning has been applied to CE in backscatter and related wireless systems to capture residual nonlinear distortions that are difficult to represent with tractable analytical models [3], [4]. Building on this capability, this paper presents a deep learning-aided phase-aware CE method for monostatic backscatter links that retains the closed-form model-based estimate as an anchor and applies a learned correction only where the linearized model becomes inaccurate, suppressing the estimation error floor caused by residual phase drift. Unlike the phase-aware lifted estimator in [2], the proposed method uses its closed-form estimate as an initialization and refines it when the first-order phase model becomes inaccurate.

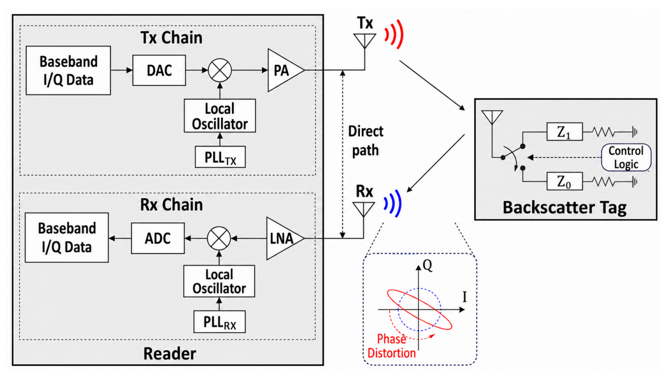

Fig. 1. The phase distortion in monostatic backscatter communications.

II. System Model

A monostatic backscatter link operating over a quasi-static channel is considered, in which the channel remains constant within each coherence frame and changes across frames. CE is performed on a pilot block per frame, where the residual intra-frame phase drift left after coarse synchronization is the dominant impairment on the recovered pilot. A phase-aware estimation model is therefore adopted, whose closed-form first-order solution remains accurate only while the accumulated drift stays within a bounded range; the deterministic error floor arising beyond this range motivates the learning-aided estimator that follows [2].

A. Residual Phase Drift in Backscatter Environment

The considered monostatic phase-distortion environment is shown in Fig. 1. The reader transmits an unmodulated carrier, the tag encodes its bits by switching between two antenna loads to impose a binary reflection on the incident wave, and the reflected field is captured by the reader receive aperture. As the tag is passive, the baseband observation traverses the cascade of a forward (reader-to-tag) hop h_{tt} and a backscatter (tag-to-reader) hop h_{tr}, so the link is governed by the composite gain

$$h \triangleq h_{tt} h_{tr}. \tag{1}$$

Under the quasi-static assumption, h remains constant over the pilot block and is treated as a single complex scalar.

Two reader-side mechanisms shape the pilot observation. Direct carrier leakage in the homodyne front-end produces a deterministic baseband offset that is suppressed prior to CE. In addition, the transmit and receive paths are driven by independently locked synthesizers whose residual locking error, even with a shared reference, leaves an uncompensated carrier-phase mismatch after coarse synchronization. This mismatch evolves nearly linearly across the pilot block and, acting multiplicatively on the two-hop observation, does not average out over the support. Let $c[m]$ denote the pilot symbols, φ the per-sample residual phase rate, and $z \sim \mathcal{CN}(0, \sigma_z^2)$ the additive receiver noise. With a pilot block of length M indexed by $\mathcal{M} \triangleq \{0, 1, \cdots, M-1\}$, the received pilot samples are modeled as

$$r[m] = hc[m]e^{j\varphi m} + z[m], \qquad m \in \mathcal{M}. \tag{2}$$

The drift over the block is summarized by the pilot phase span

$$\Phi = |\varphi|(M-1), \tag{3}$$

the total phase rotation accumulated between the first and last pilot samples, which governs the validity of the linearized model introduced next.

B. Phase-Aware Lifted Channel Estimation Model

A phase-stationary estimator that ignores φ and matches only the constant component, $\hat{h}_{\text{sc}} = \left(\sum_m c^*[m]r[m]\right) / \sum_m |c[m]|^2$, retains a deterministic phase-dependent bias that does not vanish as $\sigma_z^2 \to 0$. To remove this bias while preserving a closed form, the drift is modeled as a small linear perturbation across the block. Expanding $e^{j\varphi m} \approx 1 + j\varphi m$ and lifting the unknowns as

$$u_0 \triangleq h, \qquad u_1 \triangleq j\varphi h, \tag{4}$$

recasts (2) as an observation linear in the lifted parameters,

$$r[m] \approx (u_0 + u_1 m)c[m] + z[m]. \tag{5}$$

With $\mathbf{u} = [u_0 \; u_1]^T$, the lifted model represents the dominant residual drift component through this first-order approximation, and its closed-form phase-aware least-squares (LS) solution returns the channel estimate $\hat{h} = \hat{u}_0$ and the residual phase-rate estimate $\hat{\varphi} = \Im\{\hat{u}_1/\hat{u}_0\}$ [2].

The accuracy of (5) is limited by the term it discards. The expansion retains only the linear part of $e^{j\varphi m}$, leaving a deterministic per-sample mismatch whose magnitude grows with the squared accumulated rotation, and hence with Φ^2. Let Φ_{th} denote the practical validity boundary of the first-order lifted model. For $\Phi \leq \Phi_{\text{th}}$, this mismatch stays below the noise-induced fluctuation, so the lifted estimate remains essentially unbiased and its error decreases with pilot energy. For $\Phi > \Phi_{\text{th}}$, the discarded curvature projects through the estimator as a bias independent of σ_z^2; the CE error then saturates at a level set by Φ rather than by noise, producing an

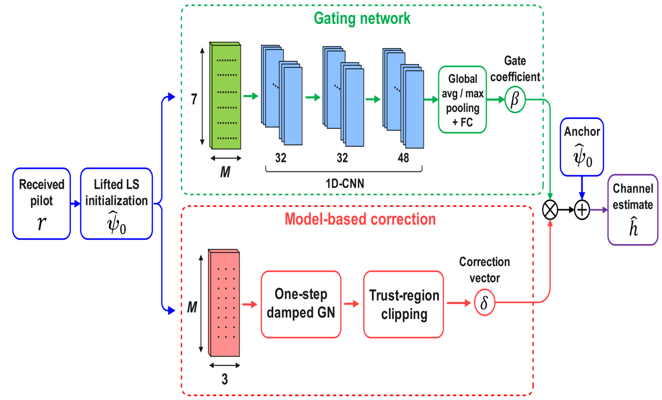

Fig. 2. The proposed RG-CNN architecture for backscatter communication CE.

SNR-independent normalized mean square error (NMSE) floor. The closed-form estimate is thus accurate and low in complexity for $\Phi \leq \Phi_{\text{th}}$, but cannot suppress the floor that arises under larger drift.

III. Deep Learning-Aided Channel Estimation

The closed-form phase-aware estimate stays accurate only for $\Phi \leq \Phi_{\text{th}}$, beyond which the discarded curvature produces a deterministic floor. A learning-aided estimator is constructed to remove this floor while retaining the model-based estimate as its operating point. The estimand is represented in real form as

$$\psi \triangleq [\Re\{h\}, \Im\{h\}, \varphi]^T \in \mathbb{R}^3, \tag{6}$$

and the lifted LS solution of (4)–(5) supplies the model-based initialization $\hat{\boldsymbol{\psi}}_0 = [\Re\{\hat{u}_0\}, \Im\{\hat{u}_0\}, \hat{\varphi}_0]^T$, with $\hat{h}_0 = \hat{u}_0$ and $\hat{\varphi}_0 = \Im\{\hat{u}_1/\hat{u}_0\}$. A residual-gated convolutional network then applies a bounded, model-based correction to $\hat{\boldsymbol{\psi}}_0$.

A. Main Network

The proposed residual-gated convolutional neural network (RG-CNN) architecture is shown in Fig. 2. Reconstructing the pilot from the initialization,

$$\hat{r}_0[m] = \hat{h}_0 e^{j\hat{\varphi}_0 m} c[m], \quad d[m] = c^*[m](r[m] - \hat{r}_0[m]), \tag{7}$$

isolates the model residual $d[m]$, which vanishes when the first-order model is exact and grows with the discarded curvature otherwise. The residual therefore carries the information not represented by the closed-form estimate, and it forms the primary input to the network.

Each pilot index contributes a real feature vector built from the matched-filtered observation $c^*[m]r[m]$, the model residual $d[m]$, the normalized index $m/(M-1)$, and the local phase coordinate $\hat{\varphi}_0 m$, with complex quantities separated into real and imaginary parts and amplitude-normalized. Stacking over m yields a feature map $\mathbf{F} \in \mathbb{R}^{7 \times M}$. Three one-dimensional convolutional layers with GELU activations process $\mathbf{F}$; dilation in the second and third layers widens the temporal receptive field across the pilot block to capture the across-block drift pattern. Global average and maximum pooling over the index axis compress the convolutional output, which is concatenated with frame-level summaries $\mathbf{s}$—the normalized initialization,

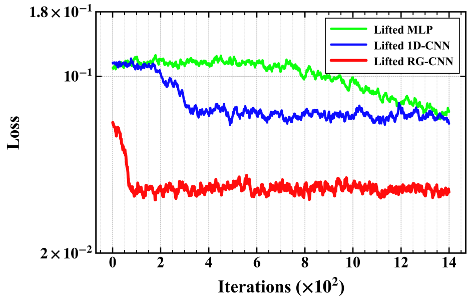

Fig. 3. Training loss convergence.

$\log|\hat{h}_0|$, the estimated span $\hat{\Phi}_0 = |\hat{\varphi}_0|(M-1)$, and residual-to-signal power ratios—and passed to a fully connected head.

B. Residual-Gated Refinement

Refinement combines a model-based correction direction with a learned gate. With $\mathbf{r} = [r[m]]_{m\in\mathcal{M}}$ and the reconstruction $\hat{\mathbf{r}}_0 = [\hat{r}_0[m]]_{m\in\mathcal{M}}$, the columns of the Jacobian of $\hat{r}_0[m]$ with respect to $(\Re\{h\}, \Im\{h\}, \varphi)$,

$$\boldsymbol{J} = \left[e^{j\hat{\varphi}_0 m}c[m], je^{j\hat{\varphi}_0 m}c[m], j\hat{h}_0 m e^{j\hat{\varphi}_0 m}c[m]\right]_{m\in\mathcal{M}}, \quad (8)$$

define a single damped Gauss–Newton (GN) step on the exact model (2),

$$\boldsymbol{\delta} = (\Re\{\boldsymbol{J}^H\boldsymbol{J}\} + \zeta\mathbf{I}_3)^{-1}\Re\{\boldsymbol{J}^H(\mathbf{r} - \hat{\mathbf{r}}_0)\}, \quad (9)$$

with damping $\zeta > 0$, clamped componentwise to a trust region $|\delta_\Re|, |\delta_\Im| \le \kappa_h$ and $|\delta_\varphi| \le \kappa_\varphi$. The fully connected head outputs a scalar gate

$$\beta = \beta_{\max}\sigma\big(f_{\boldsymbol{\Omega}}(\mathbf{F}, \mathbf{s})\big) \in [0, \beta_{\max}], \quad (10)$$

where $\sigma(\cdot)$ is the logistic function and $\boldsymbol{\Omega}$ collects the network parameters. The refined estimate follows as

$$\hat{\boldsymbol{\psi}} = \hat{\boldsymbol{\psi}}_0 + \beta\boldsymbol{\delta}, \qquad \hat{h} = \hat{\psi}_0 + j\hat{\psi}_1, \qquad \hat{\varphi} = \hat{\psi}_2. \quad (11)$$

Both the correction $\boldsymbol{\delta}$ and the gate β are driven by the same residual $\mathbf{r} - \hat{\mathbf{r}}_0$. For $\Phi \le \Phi_{\rm th}$ the residual is noise-dominated, the gate stays near zero, and $\hat{\boldsymbol{\psi}} \approx \hat{\boldsymbol{\psi}}_0$, so the closed-form accuracy is preserved. As Φ increases, the discarded curvature becomes more pronounced in the residual, and the learned coefficient β gives more weight to the exact-model correction. The trust region and the bound $\beta_{\max}$ jointly limit the departure from the model-based estimate, preventing the refinement from amplifying noise in the regime where the closed-form estimate is already accurate.

C. Training

The network is trained offline on synthetic pilots drawn from the exact model (2). For each example, $|h|$ is drawn log-uniformly and its phase drawn uniformly, the SNR is swept over a wide range, and the pilot phase span Φ is sampled across both the model-valid and large-drift regimes, so the gate is trained over both inactive and corrective regimes. Heterogeneous per-sample noise and sparse high-variance outliers are injected to discourage overfitting to a uniform-noise assumption. The gate is initialized to a small value, so optimization starts from the model-based estimate.

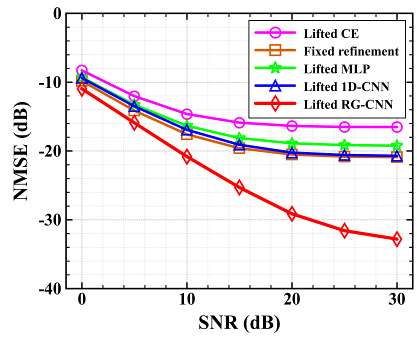

Fig. 4. CE NMSE versus SNR under residual phase drift.

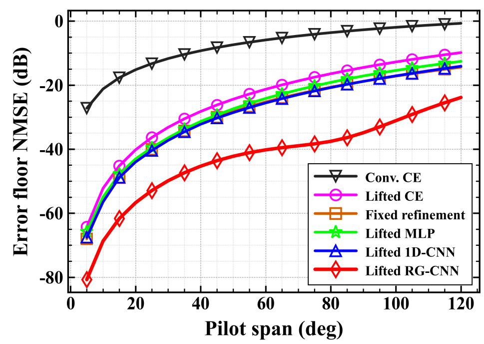

Fig. 5. Deterministic NMSE floor versus pilot phase span.

With the ground-truth parameters $\boldsymbol{\psi}$ as targets, the objective combines a channel NMSE term with auxiliary terms,

$$\mathcal{L}(\boldsymbol{\Omega}) = \mathbb{E}\left[\frac{|\hat{h} - h|^2}{|h|^2}\right] + \lambda_\varphi\mathbb{E}[(\hat{\varphi} - \varphi)^2] + \lambda_r\mathbb{E}\left[\frac{\|\hat{\mathbf{r}} - \mathbf{r}^\star\|^2}{\|\mathbf{r}^\star\|^2}\right] + \lambda_\delta\mathbb{E}\left[\|\bar{\boldsymbol{\delta}}\|^2\right], \quad (12)$$

where $\mathbf{r}^\star$ is the noiseless pilot, $\bar{\boldsymbol{\delta}}$ the trust-normalized correction, and $\lambda_\varphi, \lambda_r, \lambda_\delta$ small weights that keep the channel NMSE dominant. The reconstruction term aligns the refined estimate with the observed pilot, while the last term discourages unnecessary corrections. The network is optimized with AdamW under gradient-norm clipping, using freshly sampled mini-batches at each iteration.

IV. Numerical Results

The proposed estimator is evaluated against model-based and learned baselines, with emphasis on suppression of the deterministic error floor.

A. Simulation Settings

Performance is evaluated on synthetic pilots generated from (2). The pilot block comprises $M = 30$ antipodal BPSK symbols. The cascaded gain magnitude $|h|$ is drawn log-uniformly over $[0.02, 0.30]$ with uniform phase, the per-sample SNR spans $[0, 30]$ dB, and the residual drift is swept through a pilot phase span $\Phi \in [0°, 160°]$ covering both the

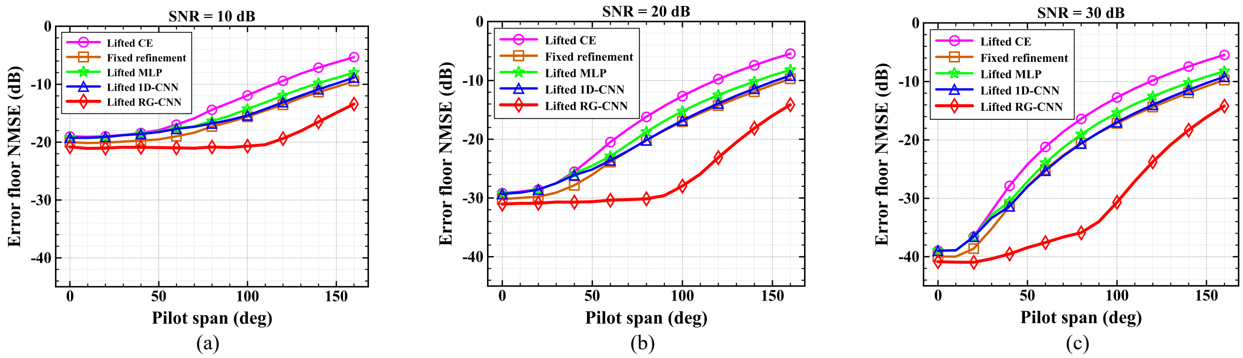

Fig. 6. Residual phase-span robustness of the CE error floor at different SNR levels: (a) 10 dB, (b) 20 dB, and (c) 30 dB.

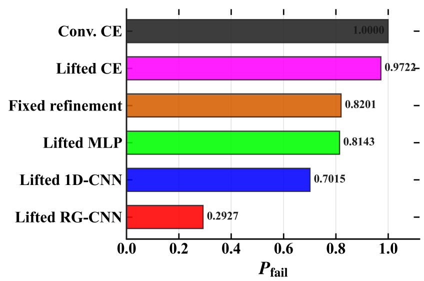

Fig. 7. Tail-failure probability comparison under large residual phase drift, where $P_{\text{fail}} = \Pr[\text{NMSE} > -20\text{ dB}]$.

model-valid and large-drift regimes. The RG backbone uses three convolutional layers (32, 32, 48 channels, kernel 5, dilation 2) and a gate head bounded by $\beta_{\max} = 0.85$, with the correction clamped to $\kappa_h = 0.25$, $\kappa_\varphi = 0.08$. Trainable parameters total $\approx 1.86 \times 10^4$, confined to the gate network, since the initialization and correction direction are model-based. Training uses AdamW over 1400 mini-batches of 256 examples drawn anew at each iteration. Evaluation samples are generated independently from the training mini-batches according to (2). The proposed RG-CNN is compared against the conventional phase-stationary estimator, the closed-form lifted estimator, a non-learned refinement, and two learned baselines, namely the lifted multilayer perceptron (MLP) and the lifted 1D-convolutional neural network (CNN), with accuracy reported as the channel NMSE and failure defined as NMSE > -20 dB.

B. Numerical Results

Training convergence is shown in Fig. 3. The proposed RG-CNN reaches its plateau within roughly 100 iterations at a loss about a factor of two below the lifted 1D-CNN and the lifted MLP baselines, as anchoring the output to the model-based estimate leaves only the bounded gate to be learned.

CE NMSE versus SNR is presented in Fig. 4. The lifted CE saturates near -16.5 dB by 30 dB, the signature of a deterministic bias that does not vanish with noise, and the fixed and learned refinements lower this plateau by only 3 to 4 dB. The proposed RG-CNN does not saturate, falling to -32.8 dB by 30 dB, a 16 dB gain over the lifted CE at high SNR.

The noiseless error floor against pilot phase span is shown in Fig. 5. Conventional CE rises from -27 dB at 5° to -0.7 dB at 120°, and the lifted CE still grows to -9.8 dB, whereas the proposed RG-CNN holds the floor at -23.9 dB—about 14 dB below the lifted CE and 23 dB below conventional CE. At vanishing span all estimators are exact, confirming that the floor is purely span-induced and that the correction removes the span bias rather than merely lowering its level.

Robustness to the residual drift is examined in Fig. 6, which plots the error floor against pilot phase span at 10, 20, and 30 dB. As SNR increases, the span-induced bias dominates earlier and the baseline curves diverge at progressively smaller spans. The RG-CNN instead maintains a nearly flat low-error region over a wider phase-span range; at 30 dB, it remains near -40 dB up to roughly 70°–80°, whereas the baselines begin to degrade below 40°, effectively extending the admissible span Φ_{th}.

The tail behavior under large residual drift is summarized in Fig. 7 by the failure probability $P_{\text{fail}} = \Pr[\text{NMSE} > -20\text{ dB}]$. Conventional CE and the lifted CE fail almost certainly, at 1.00 and 0.97, and the refinement baselines reduce P_{fail} only to the 0.72–0.82 range, whereas the proposed RG-CNN lowers it to 0.29, suppressing the heavy tail that governs link reliability.

V. Conclusion

A deep learning-aided phase-aware channel estimator for backscatter communications was developed to suppress the deterministic error floor of monostatic backscatter links under residual carrier-phase drift. The RG-CNN applies a bounded correction to the closed-form lifted estimate, acting only when the first-order model breaks down. The estimator avoided the SNR-independent saturation of the lifted estimate, lowering the channel NMSE by about 16 dB at 30 dB SNR and reducing the large-drift failure probability from near unity to 0.29.

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